Assignment Quiz # 02

National University of Computer and Emerging Sciences – Lahore Campus
Assignment 2, Computer Org. and Arch (EE), Fall 2018, Dec 5, 2018
Max Marks = 10, Time Allowed [redacted]

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Na[redacted] Registration #: [redacted]

A non-pipelined system takes 30ns to process a task. The same task can be processed in a six-stage pipeline with a clock cycle of 5 ns. Determine the speedup ratio of the pipeline for 500 tasks. What is the maximum speedup that can be achieved?

[CLO 5] [10]

30ns → 1 non-pipelined task

5ns → Pipelined task

500 tasks will take :

$= 500 \times 5ns$

$= 2.5 \mu s$

$$S_{1,k} = \frac{500 \times 5 \times 10^{-9} \times 6}{[k-1] \times 30 \times 10^{-9}}$$

~~Rough~~

~~n (time x~~

~~[k+1] time~~

$$= \frac{500 \times 5 \times 10^{-9} \times 6}{5 \times 30 \times 10^{-9}}$$

$(k = 6)$

$= \cancel{16.667}$

$= 100$

Solution:

$T_{np} = 30ns$

Tasks = 500

Time:
$500 \times 30ns = 15000ns$

↑
non-pipeline

Six-stages pipeline
$T_p = 5ns$

Tasks = 500

Time:
$= [6+(500-1)]5$
$= 505 \times 5$
$= 2525ns$

↑
pipeline

$$\text{Speedup} = \frac{15000}{2525} = 5.94$$

Using Formula:

$$\text{Speedup} = S = \frac{n\,T_{np}}{[k+(n-1)]\tau}$$

in this example, $T_{np} = k\,T_p$